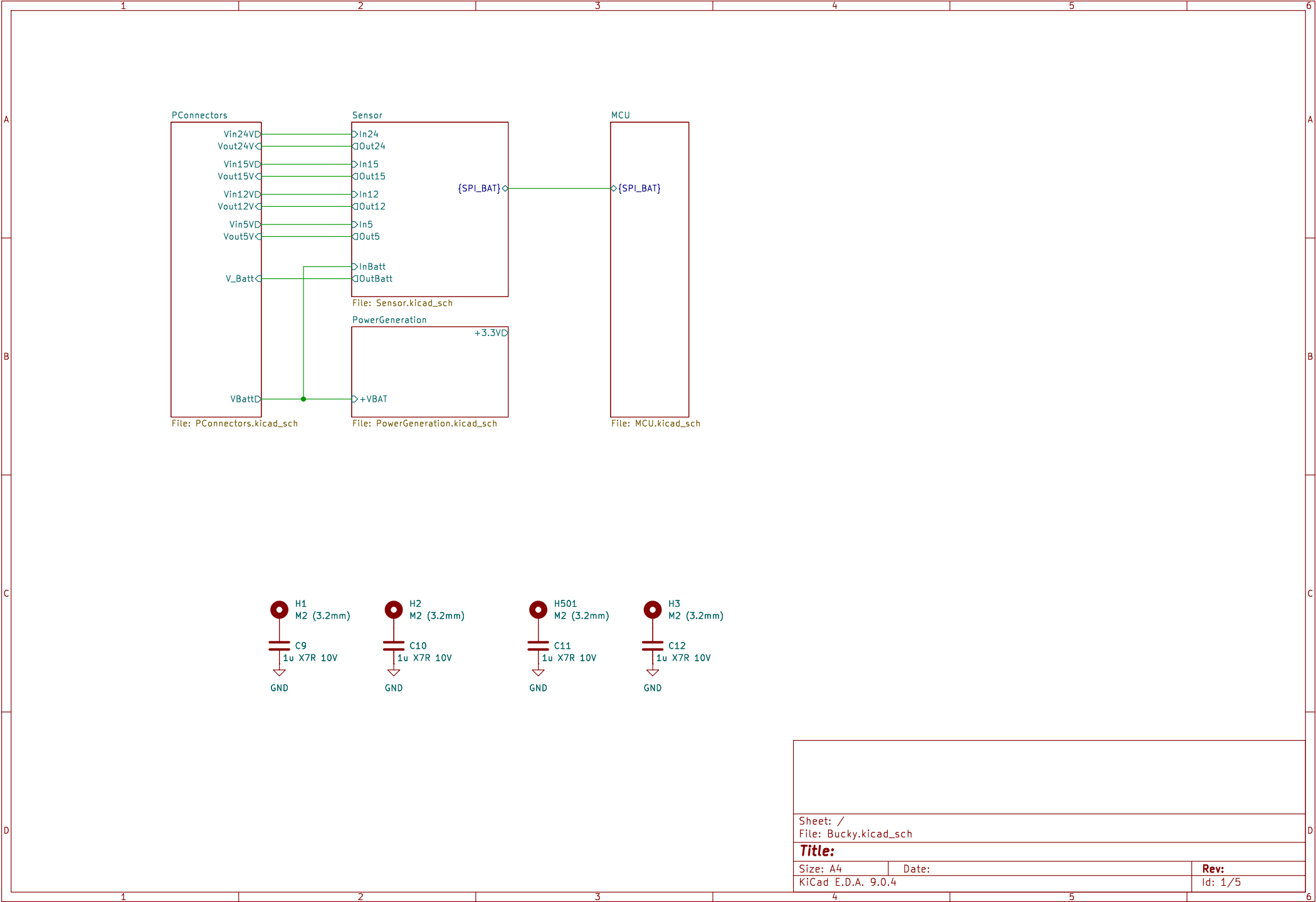
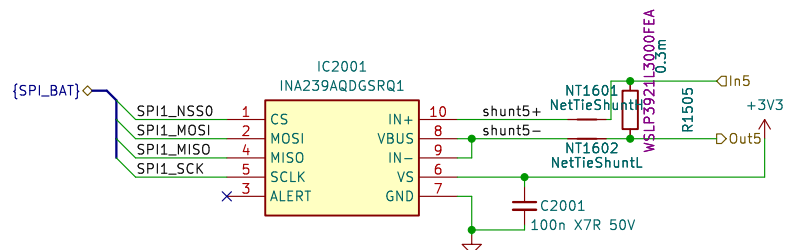
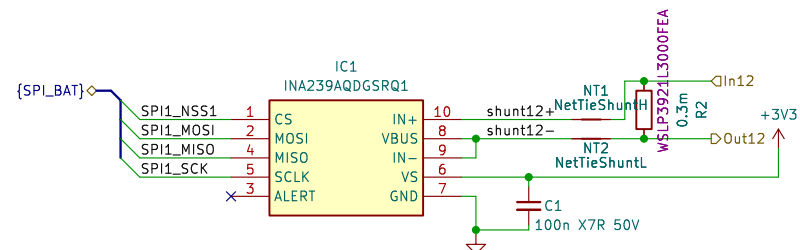




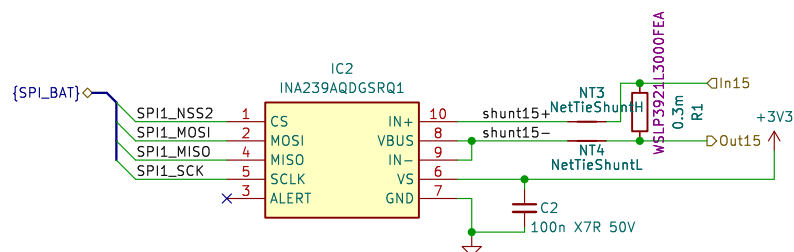
## 5V Sensor



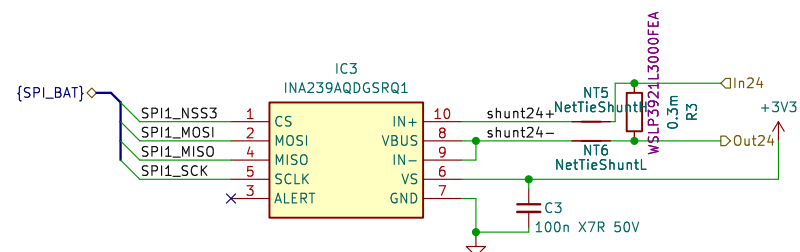
## 12V Sensor



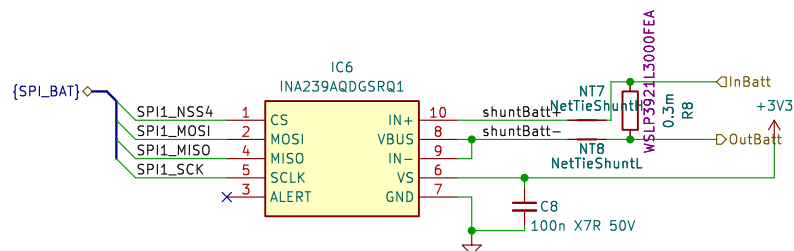
## 15V Sensor



## 24V Sensor



## Battery Sensor



Sheet: /Sensor/  
File: Sensor.kicad\_sch

### Title:

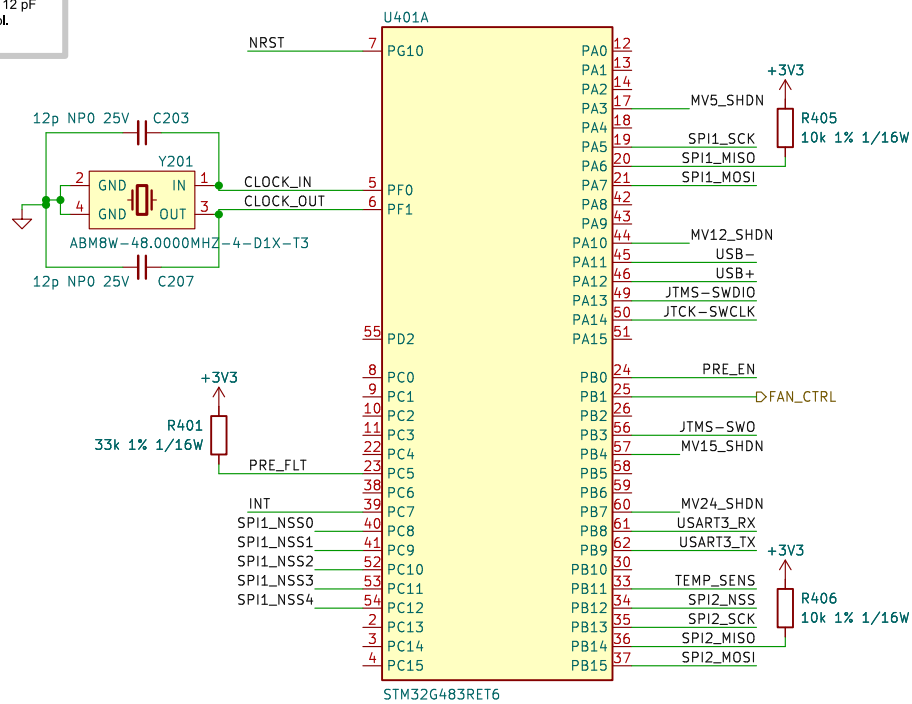
Size: A4 Date:

KiCad E.D.A. 9.0.4

Rev:

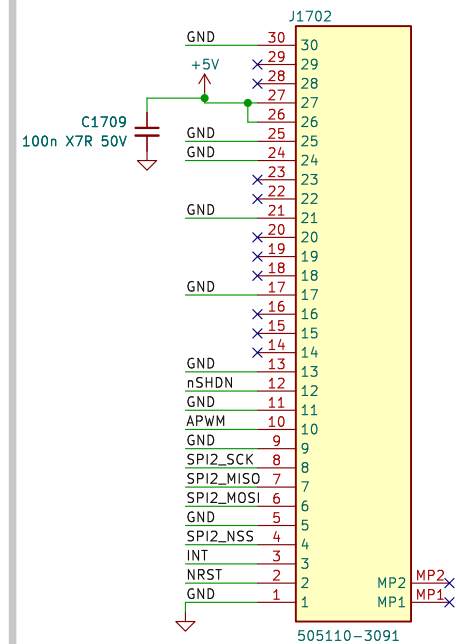
Id: 2/5

DESIGN NOTE:  
Cs = 2 pF  
CL = 8 pF  
CL1 = CL2 = 2\*(CL-CS) = 12 pF  
Choose C0G/NP0 1-2% tol.



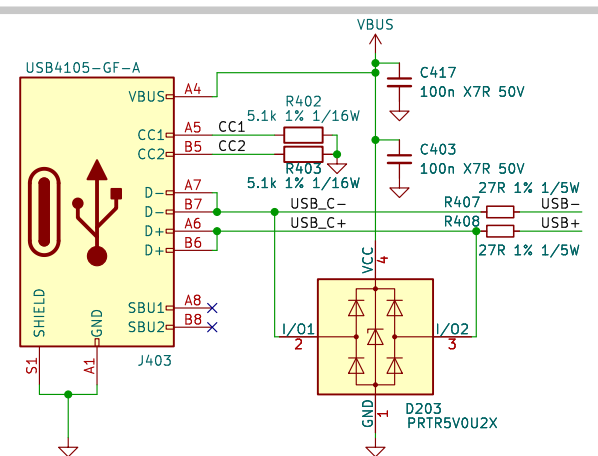
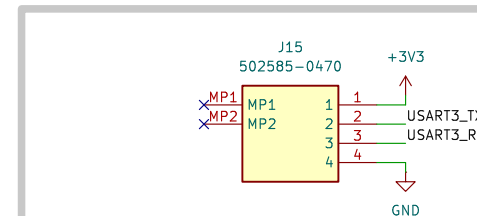
MCU I/O Domain

STM32 MCU that controls the power and communicate sensor data. The same model and same pins are used as GEODE to facilitate coding and developpement. Only the most important features are used here (Sensors, display, fan control, debug and basic communication connectors)

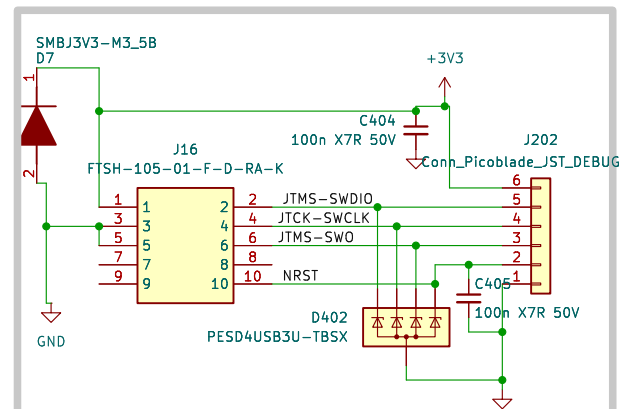


External Connector

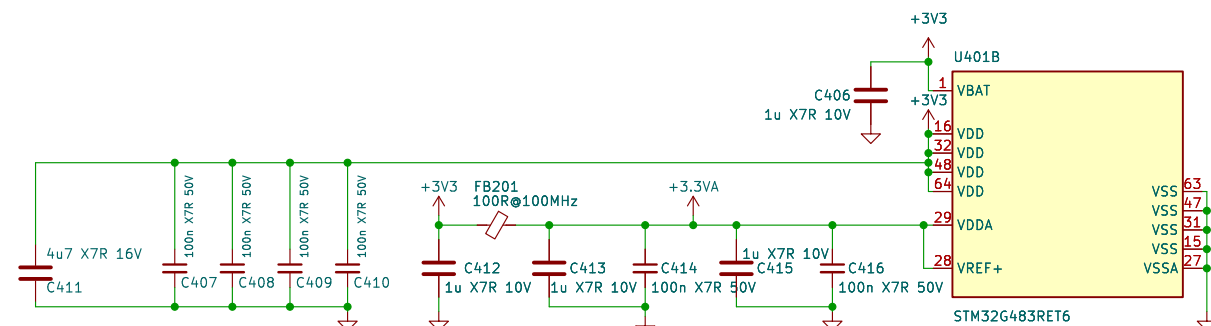
USART Connector



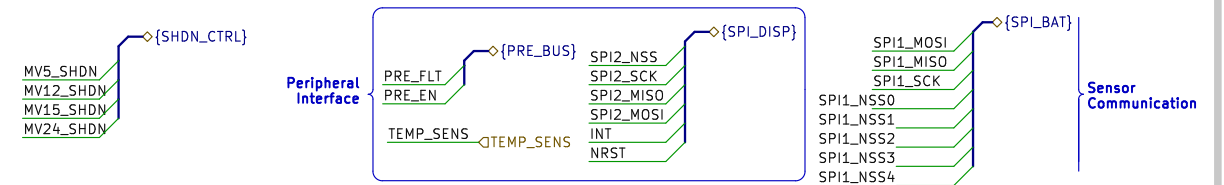
USB connector



SWD Debug Connector



MCU Power Domain



MCU Bus

Sheet: /MCU/  
File: MCU.kicad\_sch

Title:

Size: A3  
KiCad E.D.A. 9.0.4

Date:

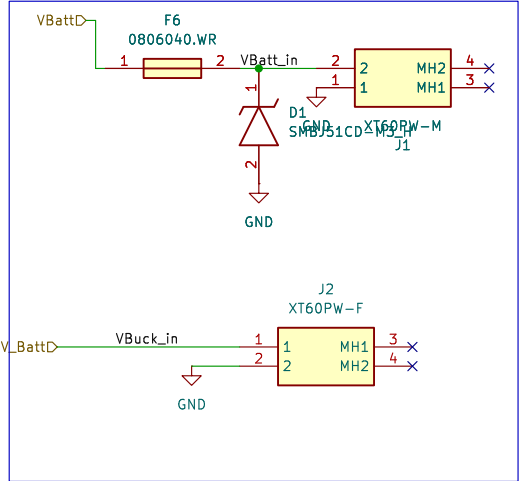
Rev:

Id: 3/5

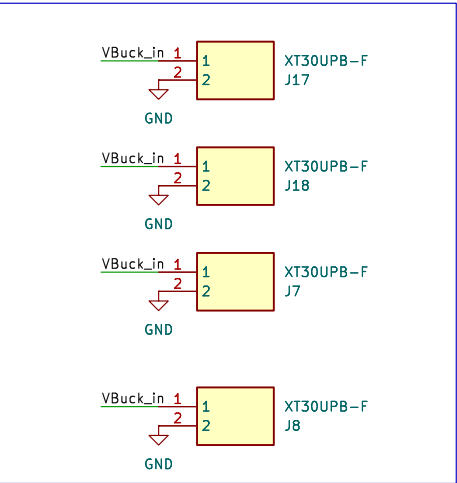
FAN

The battery is directly connected to Bucky via J1 and the power is directly transferred to the motors through J2. It is also connected to the bucks' input through VBuck\_in. The output of the buck are then connected to their respective voltage nets. The same ground is used for the whole board.

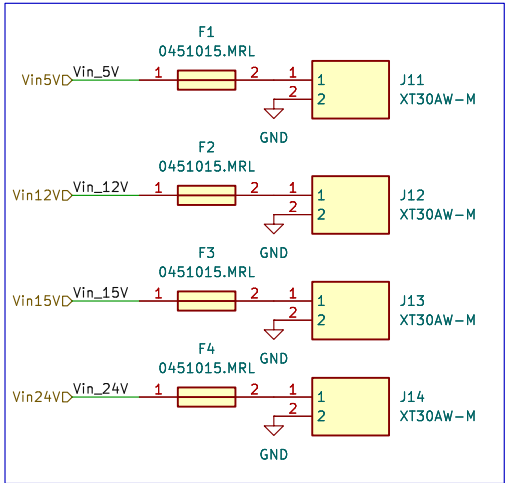
Battery Connector



To Buck Modules



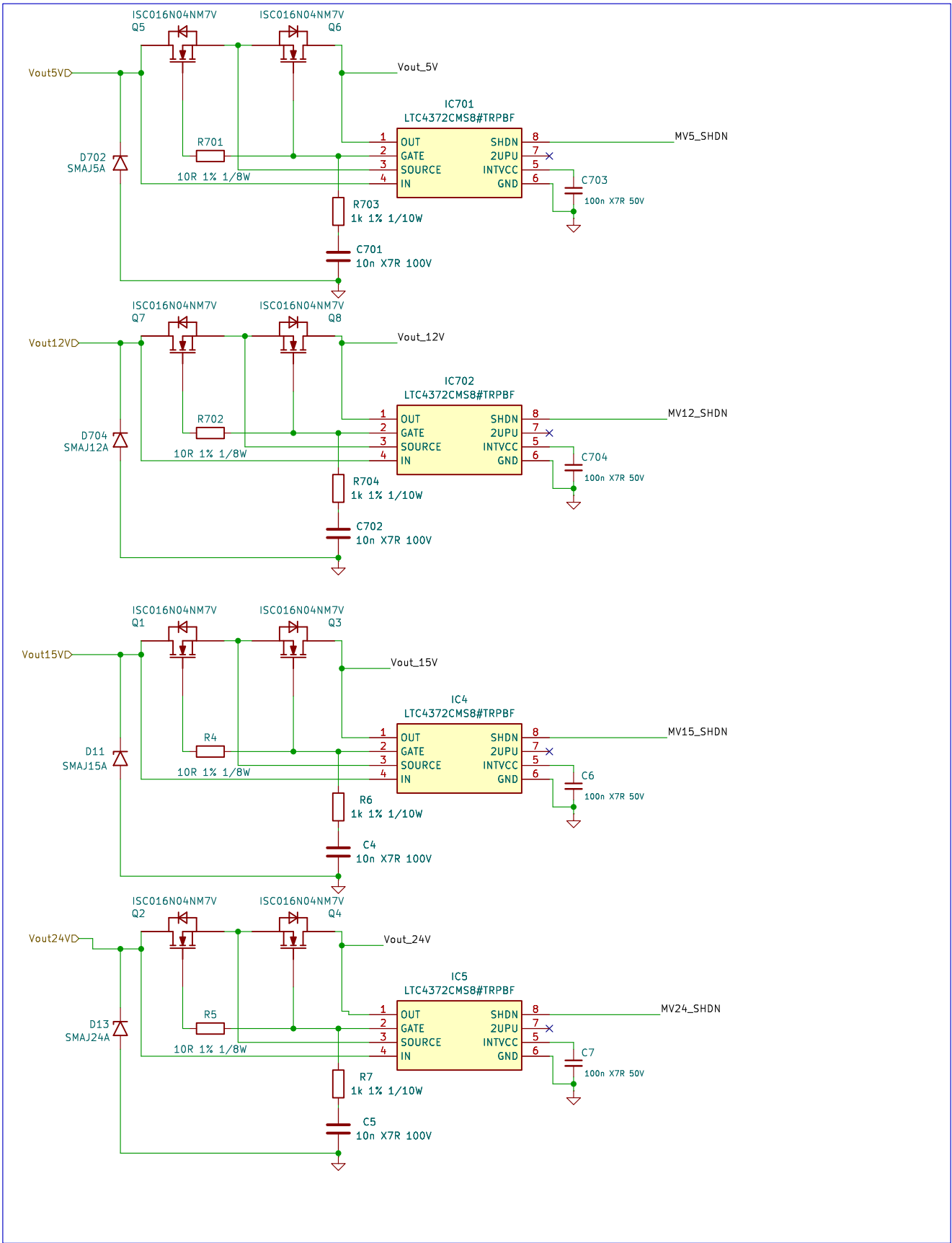
Buck Outputs



A TVS Diode is added to protect against voltage spikes after each buck output

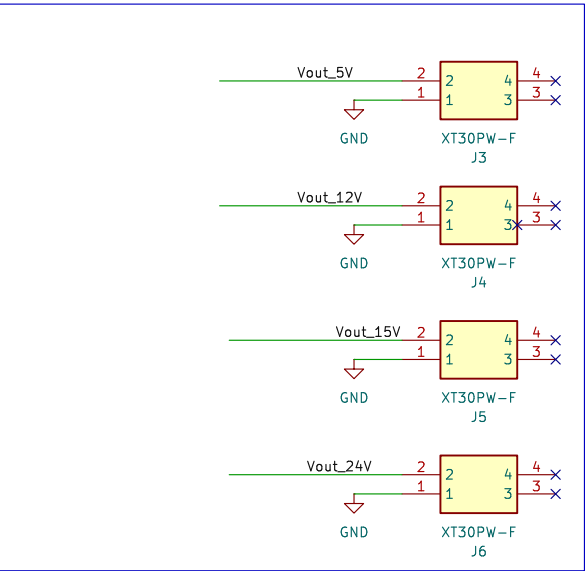
Reverse Current Protection

Reverse current protection to protect and shutdown the current in case of an issue in the bucks. Schematics adapted from GEODE

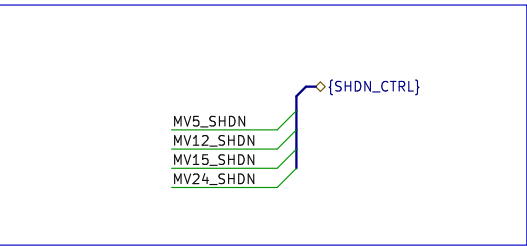


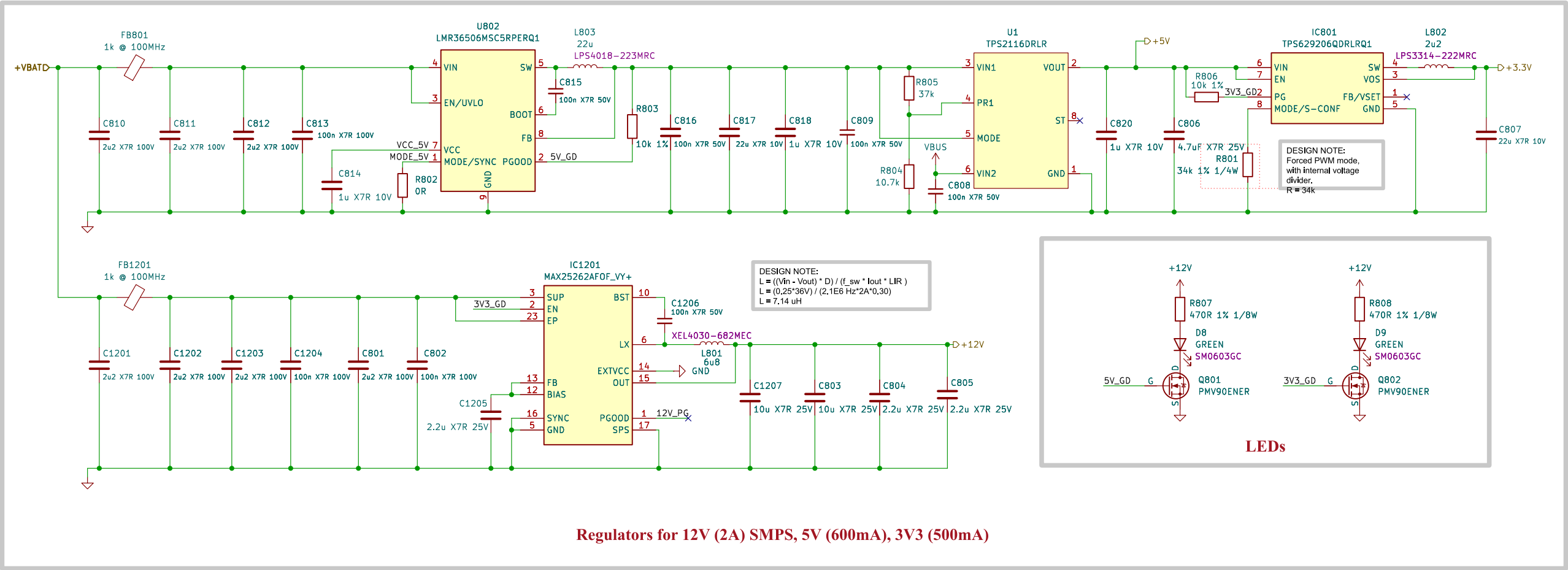
Output to the peripherals, a regular fuse is used (instead of an e-fuse) to allow for larger currents (15A)

To peripherals



Control





Local Power generation to power the MCU and peripherals. Schematics adapted from GEODE